

EGRD: ASSESSING RESEARCH IDEA NOVELTY THROUGH EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

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Paper under double-blind review

ABSTRACT

Language models now propose research ideas faster than anyone can vet them, which makes assessing novelty a practical bottleneck. Judging novelty requires identifying what prior work already covers and evaluating the contribution that remains. Overall similarity provides an incomplete account: an idea’s components may be covered across several papers, while their combination may still introduce a substantive change. We propose Evidence-Grounded Residual Decomposition (EGRD), which assesses novelty relative to an explicit set of temporally admissible prior works, making the basis of each judgment inspectable and allowing the effect of changing the evidence to be examined. EGRD estimates how much of the target those priors jointly cover, decomposes the target into latent semantic units, and learns a support matrix aligning these units with the evidence. It models two complementary residuals: unit residuals capture content with limited historical support, while interaction residuals capture unit combinations that no single prior supports. These residuals, together with coverage and contextual representations, predict an ordered novelty grade. A conservative blockwise aggregation rule lets a model trained with a small evidence budget read larger evidence sets at inference time without retraining. We further construct IDEADELTA, a dataset spanning graph machine learning, computer vision, and time series, pairing each idea with temporally admissible priors annotated for coverage, residual contribution, and novelty. EGRD recovers the annotation scheme consistently across the three domains, and ablations show that its predictions depend on the supplied evidence and on the residual representations rather than on the target idea alone. Transfer to expert judgments remains sensitive to where the grade boundaries are placed.

1 INTRODUCTION

Language models now propose research ideas faster than anyone can vet them, and deciding which of those ideas is new is the step that does not scale. Novelty is not a monotone function of textual similarity to any single prior work. We therefore study *evidence-conditioned* novelty assessment: grading a target idea against an explicit set of prior works published before it, which makes the basis of each judgment inspectable and turns the evidence into a variable we can intervene on.

Most prior work takes a paper as the unit of assessment, from citation-pattern measures (Uzzi et al., 2013) to document-level embeddings (Cohan et al., 2020; Ostendorff et al., 2022) to systems that review a submission as a whole (Zhu et al., 2025). But a paper bundles contributions of uneven novelty, and one score over the bundle does not say which contribution it describes. Systems that do operate on ideas assemble the judgment as a language-model workflow: retrieve related work, then prompt a model to compare against it (Wang et al., 2024; da Silva et al., 2025; Afzal et al., 2026). Here the evidence is selected by the system being evaluated, so no verdict can be attributed to the retrieval or to the comparison, and the comparison itself is elicited by a prompt rather than learned against supervision. Evaluating this arrangement against expert judgments, Schopf & Färber (2026) find that the reasoning such models produce mirrors human rationales, yet that alignment does not translate into accurate grades.

Such a workflow also compresses the comparison into a single score, and three distinctions do not survive. Coverage can be *dispersed*: several papers may jointly account for an idea that no single paper resembles. Ideas at equal coverage differ in *what remains*: one leaves an implementation detail unexplained, another introduces a new mechanism. And individually covered components may form a *combination* that no single prior work puts together. Each distinction asks about a particular part of the idea and the particular prior work responsible for it, and a score that discards that correspondence cannot be post-processed to recover it. Assessing novelty calls instead for a representation that keeps it.

We propose Evidence-Grounded Residual Decomposition (EGRD), which decomposes a target against ranked historical evidence into what that evidence accounts for and what it leaves, then predicts an ordered novelty grade from the two. Because existing benchmarks do not pair ideas with the evidence admissible when they were proposed, we also construct IDEADELTA to supply that pairing. Our contributions are:

- **IDEADELTA, a multi-domain evidence-paired dataset.** 21,111 ideas paired with temporally admissible priors and annotated at the idea and pair levels (92,876 pairs) across graph machine learning, computer vision, and time series. Supplied rather than retrieved, the evidence can be intervened on: permuting priors while holding targets fixed drops accuracy from 0.78 to 0.35, the majority-class band.
- **Unit-level coverage over dispersed evidence.** We model coverage at the level of latent semantic units, aligning each unit with each prior through a learned support matrix. Noisy-OR aggregation lets several partially relevant papers jointly cover a unit none covers alone; removing it costs 0.27 quadratic weighted kappa (QWK).
- **Typed residuals.** We split what the priors leave over into unit residuals (content no prior supports) and interaction residuals (combinations no single prior puts together). Both read off the same support matrix, so an unexplored contribution and an effective recombination are distinguished by the model rather than by assertion.
- **Evidence that the decomposition, not the backbone, carries the result.** EGRD reaches 0.909/0.882/0.898 QWK across the three domains against 0.579 for the best prompted system, whose scores vary more with the backbone than with the prompt chain. A block-wise rule extends the frozen checkpoint beyond its training evidence budget, leading every RINoBench (Schopf & Färber, 2026) baseline on QWK (0.2245 against 0.1469) and Macro-F1 (0.3407 against 0.2457); that margin is modest.

Appendix A situates EGRD against automated idea generation, prior novelty judges, and decomposed verification against evidence.

2 DATASET

Novelty is not a property of an idea alone; it is a property of an idea read against the work that preceded it. Evaluating it therefore requires three things for every example: a target idea, the prior ideas that were available when the target was proposed, and the contribution that survives those priors. Assembling all three together is what IDEADELTA is for.

2.1 TASK AND LABELS

The unit of annotation is a research *idea*: a single innovation claim extracted from a paper. A paper that introduces both a new objective and a new architecture therefore contributes two ideas, annotated separately. For a target idea x published at time t_x , the admissible priors are $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$. This temporal constraint does real work: an idea published after the target says nothing about the target’s novelty, however similar the two are.

IDEADELTA annotates each target at two levels. At the *pair* level, every target–prior pair receives one of four coverage levels—NOT COVERING, RELATED BUT NOT COVERING, PARTIAL COVER, LARGE COVER—recording how much of the target that single prior accounts for; a target’s retained priors also come with a ranking, which supplies the ordering a model needs when it reads evidence one prior at a time. At the *idea* level, the target receives an ordered novelty grade from LOW to HIGH.

The two levels answer different questions: pair labels say what each piece of evidence covers, the idea grade says what survives all of it. A target can hold no strongly covering prior yet still score low, when several partial covers together leave little behind.

The idea-level grade is not an overall impression layered on top of the pair labels. We instead annotate four quantities per target and combine them by the fixed rule $\text{raw}(x) = (1 - C_x) \cdot \Delta_x \cdot E_x \cdot (1 - \lambda S_x)$, $\lambda = 0.45$, where C_x is how much of the target the retained priors jointly cover, Δ_x the substantive contribution that remains, E_x whether the retrieved evidence suffices for a judgment, and S_x how much of the claimed difference is merely surface-level. Binning $\text{raw}(x)$ yields the four ordered grades. The rule lowers novelty for exactly four reasons, each separately inspectable: prior work already covers the target, the remaining difference is not substantive, the evidence is too thin to judge, or the difference amounts to renaming.

2.2 CONSTRUCTION

Paper selection and idea extraction. We collect arXiv papers in graph machine learning, computer vision, and time series, preferring work that was subsequently published and, among that, work at major venues. An LLM extracts innovation claims from each title, abstract, and introduction. A second pass discards statements that make no substantive research claim—open-source announcements, bare state-of-the-art assertions, and engineering deliverables—so that later stages grade claims rather than boilerplate.

Prior retrieval. For each target, nearest-neighbor search over idea embeddings returns candidate priors published earlier, and an LLM adjudicates the uncertain similarity band. A cross-type retrieval pass links a target to relevant prior work of a different contribution type, which same-type retrieval alone would record as isolated. At most five ranked priors are retained per target, and every pair stays within one research domain.

Annotation. A single judge, `deepseek-v4-pro`, annotates all stages. Pair coverage follows a hybrid rule: clearly covering and clearly unrelated pairs take their label directly, while the uncertain band goes to the judge, which reads the two ideas’ technical content. The judge then estimates, for each target, joint coverage, residual contribution, evidence sufficiency, and surface-level difference, and the novelty rule above converts these into the idea-level grade. The labels are model-generated; we inspected samples from every stage. Appendix B.1 gives the full pipeline.

The result is 21,111 annotated ideas and 92,876 target–prior pairs spanning 2019-01 to 2026-09, split by date rather than at random so that no training target postdates an evaluation target; Appendix B.1 reports the composition per domain. Because the labels are model-generated, we also check that they behave as novelty labels should—responding to the retrieved evidence rather than to target metadata alone, and falling when a prior already covers the target contribution—and report those checks in Appendix B.2.

3 METHODOLOGY

Given a target research idea and a set of ideas proposed before it, EGRD assesses the target’s novelty relative to the supplied evidence in three stages (Figure 1). Stage A selects temporally admissible priors and estimates how much of the target they already account for. Stage B decomposes the target against that evidence into two complementary components: unit residuals capture semantic content with insufficient historical support, and interaction residuals capture potentially novel combinations of those units. Stage C predicts an ordered novelty grade from LOW to HIGH from the residual representations. Throughout, $[\cdot; \cdot]$ denotes concatenation, $\odot$ elementwise multiplication, d the encoder dimension, σ the sigmoid, ε a small constant added for numerical stability, and $\mathcal{V}$ the non-padding positions of the target; padded priors are masked out.

3.1 STAGE A: HISTORICAL EVIDENCE INPUTS

Given a target idea x with timestamp t_x , we restrict candidate priors to those predating it, $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$, and retain the top- K ranked candidates—or all of them, if fewer are available—as

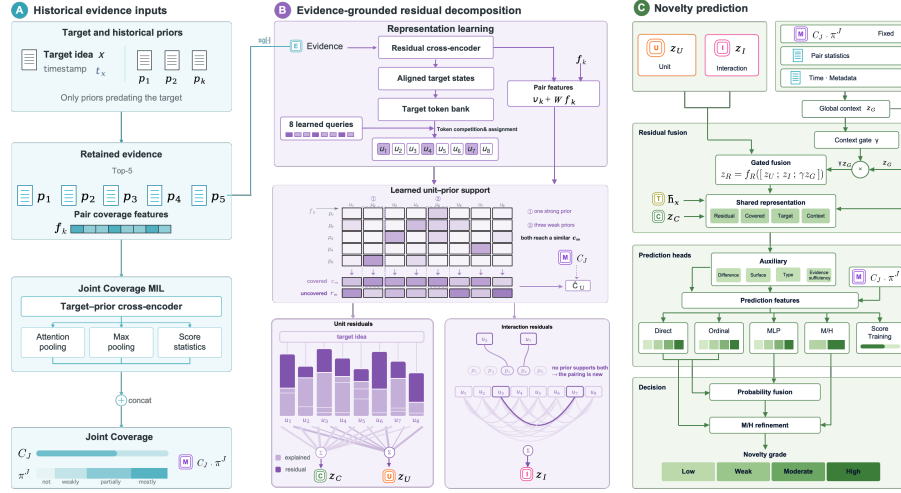


Figure 1: Overview of EGRD. (A) Only priors predating the target are admissible; the retained top- K evidence carries pair coverage features f_k , and a multiple-instance model predicts joint coverage C_J and its four-class distribution π^J . Stage A outputs enter the later stages as fixed inputs, marked $\text{sg}[\cdot]$. (B) M learned queries partition a target token bank into semantic units u_m ; a support matrix A_{km} aligns each unit with each prior, yielding the unit residual z_U , the covered pooling z_C , and—through co-support A_{ij} —the interaction residual z_I . (C) A gate modulates global context within residual fusion; the shared representation feeds three novelty branches and the moderate-to-high refinement, while the continuous score head supplies training supervision only.

the evidence set $\mathcal{E}_x \subseteq \mathcal{P}_x$ ($K = 5$ in our experiments). Each retained prior p_k carries a vector of pair coverage features $f_k \in \mathbb{R}^{11}$.

To estimate coverage across the evidence set, a cross-encoder encodes each target–prior pair and a multiple-instance model (Ilse et al., 2018) aggregates the pair representations through attention pooling and maximum pooling. Concatenated with evidence-score statistics, the pooled representations predict joint coverage C_J and a four-class coverage distribution π^J . The retained evidence, pair features, and coverage predictions serve as fixed inputs to Stages B and C, grounding the residual decomposition and providing coverage information for novelty prediction.

We summarize the evidence set in a global context vector z_G , a learned projection of twenty-one scalars—the scaled target timestamp, ten pair-score statistics over $\mathcal{E}_x$, and ten joint-coverage features—concatenated with embeddings of two target facets (aspect and contribution type). Apart from the timestamp and the facets, z_G re-encodes the Stage A outputs, and Stage C consumes it as a summary of what the evidence collectively establishes. Appendix C.2 lists the pair and context features and specifies the pair-scoring rule and the joint model.

3.2 STAGE B: EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

Stage B decomposes the target into semantic units and estimates their support from the retained evidence, representing the unexplained contribution through unit residuals for content with limited historical support and interaction residuals for combinations with limited joint support from any single prior.

Semantic decomposition. A residual cross-encoder, initialized from the Stage A model and trained separately from it, produces a pooled representation v_k for each target–prior pair, which we augment with projected pair coverage features as $\tilde{v}_k = v_k + f_P(f_k)$. Averaging the contextual token states at matching positions across valid pairs yields a shared token bank $\{h_j\}_{j=1}^N$ over the N target tokens. We then use M learned queries q_m to softly assign these tokens to semantic units (Locatello et al., 2020) ($M = 8$ in our experiments): the queries compete for each token, $B_{mj} = \text{softmax}_m(q_m^\top h_j / \sqrt{d})$, and the resulting assignments are normalized over tokens within each unit, $\alpha_{mj} = B_{mj} / (\sum_{j' \in \mathcal{V}} B_{mj'} + \epsilon)$, giving $u_m = \sum_{j \in \mathcal{V}} \alpha_{mj} h_j$. The units have

no predefined semantic roles. A learned activity score $\phi_m = \sigma(\mathbf{w}_\phi^\top \mathbf{u}_m + b_\phi)$ controls each unit's contribution, while an orthogonality penalty on the attention matrix discourages redundant token assignments.

Evidence-aligned unit residuals. A learned support matrix estimates how strongly each prior p_k supports each unit $\mathbf{u}_m$, $A_{km} = \sigma(f_A([\mathbf{u}_m; \tilde{\mathbf{v}}_k; \mathbf{u}_m \odot \tilde{\mathbf{v}}_k; |\mathbf{u}_m - \tilde{\mathbf{v}}_k|; \mathbf{f}_k]))$, where f_A is an MLP. Support is estimated separately for every target–prior pair, so different priors can account for different units. Over the $n = |\mathcal{E}_x|$ retained priors, noisy-OR aggregates support into unit coverage $c_m = 1 - \prod_{k=1}^n (1 - A_{km})$; its complement, weighted by unit activity, defines the residual weight $r_m = \phi_m(1 - c_m)$ and the unit residual $\mathbf{z}_U = M^{-1} \sum_m r_m \mathbf{u}_m$. Noisy-OR lets several partially relevant priors jointly cover a unit that none of them covers alone; we use it as a differentiable aggregation rule, not as a claim that the priors are independent sources of support (Appendix C.3). An active unit with little historical support therefore receives a large residual weight.

Throughout Stage B we preserve the magnitude of the residuals rather than normalize it away. We divide $\mathbf{z}_U$ by the fixed number of units M rather than by $\sum_m r_m$, so that the pooled representation retains the total residual mass; the supported content is pooled the same way, as $\mathbf{z}_C = M^{-1} \sum_m \phi_m c_m \mathbf{u}_m$. To connect unit coverage to supervision, a learned, activity-weighted pooling of c_m reconstructs joint coverage $\hat{C}_U$, supervised by annotated coverage and the fixed Stage A prediction. The support entries themselves are latent strengths learned without unit-level alignment labels.

Interaction residuals. Units that are individually covered may still form a new combination. For each unit pair $i < j$, an MLP maps their concatenation, elementwise product, and absolute difference to an interaction representation $\mathbf{t}_{ij}$, and a sigmoid head predicts its strength g_{ij} . We measure historical co-support by the largest product of the two units' support strengths within a single prior, $\bar{A}_{ij} = \max_k A_{ki} A_{kj}$, which gives the interaction residual weight $r_{ij} = g_{ij}(1 - \bar{A}_{ij})$ and the pooled representation $\mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}$, where the learned attention η_{ij} includes a $\log(r_{ij} + \epsilon)$ bias that favors pairs with larger residual weights. When different priors support the two units but no single prior strongly supports both, their unit residuals can be small while their interaction residual remains large, provided that g_{ij} is high. Absent co-support is therefore read as an absence of evidence that the combination already exists, which is what a novel combination looks like in the historical record. Co-support only proxies for that evidence: a prior that supports both units need not cover the specific way the target combines them. The pooling keeps the residual magnitude here as well— r_{ij} multiplies $\mathbf{t}_{ij}$ after the attention weights are normalized—and an auxiliary classifier on $\mathbf{z}_I$ is trained with idea-level labels for nontrivial combinations (Appendix C.3).

3.3 STAGE C: NOVELTY PREDICTION

Stage C predicts an ordered novelty grade from the residual representations and the evidence context. Because evidence availability and uncertainty vary across targets, a learned gate γ scales the global context within the residual branch: the gate reads $\mathbf{z}_U, \mathbf{z}_I, \mathbf{z}_G$ and the entropy of the Stage A coverage prediction, and fusion yields $\mathbf{z}_R = f_R([\mathbf{z}_U; \mathbf{z}_I; \gamma \mathbf{z}_G])$. Layer normalization over the fused residual, the covered content, the target representation $\mathbf{h}_x$, and the context then gives the shared representation $\mathbf{h} = \text{LayerNorm}([\mathbf{z}_R; \mathbf{z}_C; \mathbf{h}_x; \mathbf{z}_G])$. Since $\mathbf{z}_G$ also enters $\mathbf{h}$ directly, the gate controls context only within residual fusion.

Auxiliary heads predict substantive difference, surface change, evidence sufficiency, and residual type. These predictions, together with $\mathbf{h}$ and joint coverage, feed three novelty branches: a direct linear classifier, a cumulative ordinal head that models grade order, and a two-layer MLP classifier. Each branch produces a four-class distribution, and their convex mixture determines an initial grade.

MODERATE and HIGH account for most of the confusion between grades, so a moderate-to-high refinement revisits that boundary once the initial grade is set. The refinement upgrades an initial MODERATE prediction when a weighted combination of the direct and ordinal HIGH-class probabilities and a separate binary-head score exceeds a threshold. We select the mixture weights and the refinement threshold on development data and fix them for testing (Appendix C.4). A separate continuous novelty head supplies auxiliary supervision during training and stays out of the inference-time grading rule.

Table 1: Baselines and our model on the IdeaDelta evaluation set, 3,352 ideas, same labels and the same idea-only protocol (top-5 retrieved priors, no paper text), split by domain. A row group isolates the effect of the model behind an unchanged prompt chain. A majority-class predictor scores 0.312, 0.356 and 0.328 accuracy with kappa zero by construction, and several rows fall below it; MAE is lowest for rows that concentrate near the middle of the scale, so read it beside QWK.

System	Backbone	Graph ML				Computer vision				Time series			
		Acc	Ma-F1	QWK	MAE ↓	Acc	Ma-F1	QWK	MAE ↓	Acc	Ma-F1	QWK	MAE ↓
Direct, 5 priors	Qwen3.5	0.300	0.266	0.319	0.925	0.308	0.264	0.300	0.876	0.291	0.283	0.327	0.956
	deepseek-flash	0.438	0.444	0.558	0.706	0.409	0.418	0.521	0.728	0.416	0.415	0.560	0.716
	gpt-5.6-luna	0.432	0.439	0.579	0.657	0.420	0.430	0.561	0.661	0.445	0.436	0.576	0.653
RINoBench	Qwen3.5	0.324	0.286	0.335	0.831	0.346	0.293	0.338	0.777	0.344	0.333	0.413	0.793
	deepseek-flash	0.443	0.458	0.560	0.639	0.416	0.411	0.501	0.676	0.442	0.447	0.553	0.654
	gpt-5.6-luna	0.391	0.377	0.487	0.698	0.393	0.364	0.458	0.683	0.392	0.379	0.512	0.688
UKPLab 3-stage	Qwen3.5	0.280	0.161	0.125	0.928	0.289	0.150	0.106	0.873	0.274	0.180	0.132	0.973
	deepseek-flash	0.351	0.267	0.301	0.761	0.376	0.272	0.259	0.741	0.386	0.289	0.290	0.727
	gpt-5.6-luna	0.318	0.226	0.266	0.800	0.367	0.248	0.304	0.721	0.365	0.267	0.296	0.743
GraphMind	Qwen3.5	0.264	0.205	0.154	1.027	0.264	0.182	0.108	0.959	0.244	0.204	0.155	1.051
	deepseek-flash	0.291	0.227	−0.007	1.095	0.342	0.266	0.154	0.971	0.365	0.249	0.159	0.916
	gpt-5.6-luna	0.289	0.255	0.046	1.003	0.381	0.328	0.304	0.755	0.380	0.327	0.305	0.750
DeepReviewer	DeepReviewer-7B	0.294	0.237	0.034	0.965	0.325	0.235	−0.043	0.972	0.296	0.222	−0.017	0.964
EGRD (ours)	—	0.799	0.801	0.909	0.201	0.761	0.767	0.882	0.242	0.789	0.776	0.898	0.215

Training. We initialize the residual encoder from the Stage A joint-coverage checkpoint and jointly optimize it with the decomposition and prediction modules, while keeping the Stage A models fixed. The objective combines classification of the novelty grade and the residual type, smooth- L_1 supervision of five annotated scalars—substantive difference, surface change, evidence sufficiency, the reconstructed coverage $\hat{C}_U$, and a continuous novelty score—and regularization of the decomposition against redundant token assignments, unused units, and coinciding unit coverages. Because level losses alone leave the scalar heads free to settle on the median of their target, each is also supervised in rank; the coverage term additionally penalizes disagreement with the fixed Stage A prediction, weighted down where that prediction has higher entropy. Appendix C.5 states the full objective, defines each term, and reports the coefficients.

4 EXPERIMENTS

The evaluation set is the IDEADELTA test split. Every system receives the same evidence — the five highest-ranked admissible priors per target, as titles and abstracts, with no paper text — and is scored against the same labels by the same script, so a difference between two rows is a difference in how the evidence is used. On a four-level label with unequal support, accuracy rewards the majority class and MAE the middle of the scale, while quadratic weighted kappa is what a constant predictor cannot win, so we report all four. Appendix D.1 gives the splits, hyperparameters and scoring code.

4.1 MAIN RESULTS

We compare against a *direct* single call, the *RINoBench* (Schopf & Färber, 2026) and *UKPLab 3-stage* (Afzal et al., 2026) prompt chains, *GraphMind* (da Silva et al., 2025), which labels each prior as supporting or contrasting before rating, and *DeepReviewer-7B* (Zhu et al., 2025), a reviewer model fine-tuned on paper review. The first four appear once per backbone. Table 1 reports every system on the IDEADELTA evaluation set, and Table 2 carries the same frozen checkpoint to RINoBench. In the first table the prompted systems occupy a band well below our row.

Among the prompted systems, the backbone matters more than the prompt chain. Holding the prompt fixed and changing the model moves the result further than the reverse. On Graph ML a single five-prior call rises from 0.300 to 0.438 accuracy when Qwen3.5 is replaced by deepseek-flash, a larger move than separates any two prompt chains on one backbone. GraphMind is the exception, and in both directions: it gains the least from a better backbone, and in all nine cells it scores below a single direct call on the backbone it runs on — on Qwen3.5, 0.264 against 0.300

Table 2: RINoBench test, 277 ideas, human expert 1–5 labels folded to four levels with 4/5 merged. The merged top class holds 41.5% of the ideas, so predicting it everywhere scores 0.4152 accuracy, at or above every system in the table. Only about 30 of the 277 ideas fall in the three fields EGRD was trained on. Bold marks the best value in a column; Acc and MAE are not led by our row.

System	Acc	Ma-F1	QWK	MAE ↓
<i>Same Qwen3.5 backbone, zero-shot</i>				
Qwen3.5-397B-A17B	0.3574	0.2244	0.1451	0.7726
GraphMind	0.3357	0.2281	0.1469	0.7653
UKPLab 3-stage	0.3466	0.1892	0.0808	0.7220
<i>Released predictions of the benchmark authors, folded to four levels</i>				
GPT-5	0.3646	0.2457	0.0955	0.8051
o3	0.3971	0.2370	0.0606	0.8087
gpt-oss-120b	0.4152	0.2249	0.0392	0.8375
Llama-3.1-8B-Instruct	0.3755	0.1961	−0.0011	0.8773
DeepSeek-R1	0.4224	0.1873	0.0283	0.8700
Llama-4-Scout-17B-16E	0.4116	0.1864	0.0215	0.8809
Llama-3.3-70B-Instruct	0.4188	0.1531	0.0051	0.9025
<i>Reviewer model fine-tuned on paper review, local weights</i>				
DeepReviewer-7B	0.1913	0.1425	0.0145	1.3610
EGRD (ours)	0.4152	0.3407	0.2245	0.8087

on Graph ML, at 476 seconds and 31,201 LLM calls for the 3,352 ideas. Its polarity stage is why: given the same five priors Qwen3.5 calls 4.33 supporting against deepseek-flash’s 1.04, and the ratings follow. The stage is also unstable across backbones, and kappa records it: on Graph ML deepseek-flash reaches −0.007, no ordering at all, while the same chain on the same backbone reaches 0.159 on time series.

The domain, by contrast, does not matter. Within any row the three domains differ by under ten accuracy points while the gap to EGRD exceeds thirty on every metric, so what separates the two blocks is not a property of one field.

Fine-tuning on paper review does not transfer to ideas. DeepReviewer-7B is the one system trained rather than prompted, and its kappa of 0.034, −0.043 and −0.017 says it does not order the ideas. A model trained to review whole papers appears to read a longer related-work block as evidence of less novelty largely independently of what that work says, which is what scoring a target against a retrieved set read as a block permits.

We evaluate the frozen checkpoint on RINoBench (Schopf & Färber, 2026), whose 277 test ideas carry expert 1–5 labels folded to four levels. These experts never saw the scheme, and the drop between the two tables is the cost of changing annotator.

The ordering transfers even though the level does not. EGRD leads Macro-F1 and kappa at 0.3407 and 0.2245 against 0.2457 and 0.1469 for the best other system on each, and is the only row above 0.15 kappa, while six of the eleven baselines sit below 0.05 and so do not order the ideas at all. Accuracy carries no information here, since a constant predictor scores 0.4152; our row reaches the same 0.4152 with a prediction that is not constant, 16/36/76/149 ideas across the four classes against a gold 15/60/87/115. The margin is modest: we read it as some agreement between residual structure learned on model-generated labels and expert judgment, not as the problem solved.

Two properties of the transfer bound what that agreement can be, and both are visible before any model is run. The first is coverage. IDEADELTA is built from three fields, whereas RINoBench draws its ideas from OpenReview submissions across machine learning as a whole, so most of its test ideas sit in areas — robust reinforcement learning, tabular learning, computational biology — whose prior work the model was never trained to read. The second is granularity. The expert scale has five levels against the four EGRD predicts, so the top two are merged, and that merge does two things at once: it puts 41.5% of the ideas into a single class, which is why accuracy is uninformative here, and it erases a distinction the experts themselves chose to draw, so a model that orders the ideas correctly still has no way to express the difference between a 4 and a 5.

Table 3: Ablations on the pooled IDEADELTA evaluation set, 3,352 ideas. **Full** is the unablated checkpoint of Table 1. The pipeline-stage rows intervene on that trained checkpoint at inference and the decomposition-design rows retrain the model without the component, so magnitudes are comparable within a block but not across the two.

System	Acc	Ma-F1	QWK	High-F1
Full	0.7796	0.7790	0.8958	0.7708
<i>Pipeline stages</i>				
Permuting prior evidence	0.3471	0.3532	0.2241	0.4420
Permuting coverage	0.6649	0.6605	0.5996	0.6630
Zeroing residuals	0.6371	0.6554	0.7878	0.7079
<i>Decomposition design</i>				
Collapsing semantic units	0.6580	0.6292	0.8591	0.6120
Disabling slot competition	0.6701	0.6874	0.7665	0.6813
Removing noisy-OR support	0.6210	0.5520	0.6291	0.5665

The blockwise rule lets the model read more evidence than training supplied. RINoBench offers far more candidate priors than the five EGRD was trained with. Splitting them into disjoint blocks of five and taking the minimum substantive delta over blocks is conservative — any block finding the idea covered lowers its grade — and needs no retraining, since a block has the shape seen in training. Raising the budget from 25 to 60 priors moves accuracy from 0.4116 to 0.4152 and kappa from 0.2182 to 0.2245: real but small, and saturating, since the median idea carries 23 candidates. The same change separates the systems: Qwen3.5 improves from 0.2244 to 0.2477 Macro-F1 while DeepReviewer-7B falls from 0.1425 to 0.1230. A system that treats the amount of retrieved prior work as the signal degrades when given more of it.

4.2 ABLATION

Because the evidence is supplied rather than retrieved, it can be intervened on with the target held fixed, which separates a model that reads the evidence from one that recognises gradable phrasing.

Every stage of the pipeline is load-bearing, and error accumulates along it. The first three interventions cut the chain at successive points. **Permuting prior evidence** shuffles the retrieved prior texts across targets, destroying the semantic content the chain is built on; **permuting coverage** shuffles only the joint coverage score and leaves every prior text in place, destroying what the historical evidence inputs (§3.1) infer from that content; **zeroing residuals** sets z_U , z_I and z_C to zero, destroying what the evidence-grounded residual decomposition (§3.2) adds on top of both. The ablations show that all three are indispensable. Permuting prior evidence costs 0.43 accuracy and 0.67 kappa, dropping accuracy into the band of the majority-class predictor of Table 1 (0.312–0.356): the grade is not recoverable from the target alone, so the priors are read rather than treated as a pretext for grading the target’s own phrasing. Permuting coverage costs 0.11 accuracy and 0.30 kappa with the texts intact, so what the historical evidence inputs infer is not redundant with the evidence it is computed from. Zeroing residuals costs 0.14 accuracy, 0.12 Macro-F1 and 0.11 kappa, so the residual decomposition is not an ornament on a prediction the evidence inputs already support. Taken in order, the damage grows the earlier the cut falls: what one stage corrupts it passes on, and the corruption is amplified rather than absorbed downstream. Accuracy alone reverses at the last step, 0.14 for zeroing residuals against 0.11 for permuting coverage, and it reverses for a reason: the residual decomposition moves the metric that counts exact grades far more than the one that counts agreement in rank, so what it contributes is which grade an idea receives rather than how the ideas are ordered.

The mechanisms inside the residual decomposition earn their place, and two of the three act on the ordering. The remaining interventions leave the pipeline intact and remove one mechanism from the decomposition itself. **Collapsing semantic units**, from eight to one, costs 0.12 accuracy, 0.15 Macro-F1 and 0.16 High-F1 but only 0.04 kappa, so the collapse loses the grade and leaves the ordering nearly intact. **Disabling slot competition**, which removes the softmax through which the units compete for each token, costs 0.11 accuracy and 0.13 kappa, reversing that pattern: deciding which unit a token belongs to is what keeps the units apart in the first place. **Removing noisy-OR support** costs the most of the three on every metric, 0.16 accuracy, 0.23 Macro-F1, 0.27 kappa and 0.20 High-F1, so aggregating support across priors is what makes the ordering work. For the

collapsed units and the removed noisy-OR, High-F1 falls further than accuracy, which is where a decomposition should matter: the top grade turns on parts of a target that no single prior supports.

We also sweep the three parameters that govern how much the model is given (the evidence budget K , the training fraction and the encoder input length) one at a time from the default configuration of Table 1, in all three domains. The largest of the three moves accuracy by at most 0.075, against the 0.34 that separates our row from the best prompted row of Table 1, so none of them is what the main comparison turns on. The sweeps, and what they do and do not license, are in Appendix D.2.

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A RELATED WORK

Automated research and idea generation. Language models are increasingly used to carry out parts of the research process itself. Wang et al. (2024) retrieve inspirations from past literature and optimise explicitly for novelty by comparing a candidate against prior papers; Baek et al. (2025) refine proposals against a set of LLM reviewing agents that give structured feedback along several dimensions; and Lu et al. (2024) close the loop from ideation through experiments to a written

paper and a simulated review. Si et al. (2024) evaluate what such pipelines produce, finding the generated ideas rated more novel than human-written ones but less feasible. Every one of these systems contains an internal novelty check, and it is the check rather than the generator that bounds them: once proposals are cheap, the output of the pipeline is whatever its filter admits. The filter is nonetheless the component these systems evaluate least directly, judged only through the ideas that eventually emerge downstream of it.

Judging the novelty of scientific work. Studied on its own, the check has passed through several generations of method. The earliest operate on citation statistics, measuring how unusual the combinations of references in a paper are against historical co-occurrence (Uzzi et al., 2013); they require the work to already exist and be cited, and reduce novelty to patterns of co-occurrence rather than to what a paper says. Document-level semantic representations replace surface statistics with learned relatedness, scoring how close a submission is to prior papers through citation-informed embeddings (Cohan et al., 2020; Ostendorff et al., 2022), but they compare documents as points and recover nothing about which content is shared. Most recent systems retrieve related literature and hand the comparison to a language model. Liang et al. (2024) obtain structured feedback from a single prompt, of which significance and novelty is one section, and report partial overlap with human reviews; Zhu et al. (2025) train a reviewer model on a multi-stage process of structured analysis, retrieval, and evidence-based argumentation; Afzal et al. (2026) split the judgment into claim extraction, retrieval and synthesis of related work, and structured comparison, motivating the design with an analysis of human-written novelty reviews; and da Silva et al. (2025) build an interactive system on the same retrieve-then-compare pattern. Schopf & Färber (2026) supply the evaluation counterpart and report that model ratings still diverge substantially from expert ones even when the accompanying reasoning appears sound. Two properties persist across these generations, and both are structural rather than incidental. The evidence is retrieved by the system under evaluation, so its verdicts are unattributable: an error cannot be assigned to the retrieval or to the judgment, and the two therefore cannot be improved, or even measured, separately. We remove this by conditioning on a given, temporally admissible evidence set, which additionally turns the evidence into a variable that can be intervened on. The second property is that the target is scored as an undivided whole, against the closest prior work or against a retrieved set read as a block. Such a comparison returns one number per target, and no post-processing of that number can recover which part of the idea it is about; the information is discarded before the score is formed, not lost afterwards.

Decomposed verification against evidence. Saying which part remains requires decomposing the target, and outside novelty assessment this has become the standard recipe for factuality evaluation: break a text into smaller propositions and check each against retrieved evidence. Min et al. (2023) decompose a passage into atomic facts, verify each against a knowledge source, and report the supported fraction; Wei et al. (2024) scale the same procedure with search as the evidence source; and in the scientific domain the corresponding task checks a stated claim against candidate abstracts while identifying the supporting rationales (Wadden et al., 2020; 2022). Three properties of the recipe do not survive the move to research ideas. The propositions are assumed to be given, whereas an idea carries no stated claims, and the decomposition step is not a neutral preprocessor: Wanner et al. (2024) show that scores of this family shift materially with the decomposition used. Support is treated as a label, whereas a prior work can cover part of a unit and not the rest. And each unit is verified independently before the results are averaged, so the score is a function of the per-unit coverages alone and cannot express whether a combination of units has precedent. We accordingly induce the units rather than reading them off the text, measure support as a graded quantity, and add a residual defined over pairs of units.

B DATASET DETAILS

B.1 DATASET CONSTRUCTION PIPELINE

Sources and idea extraction. Papers are drawn from arXiv in three areas—graph machine learning, computer vision, and time series. Selection prefers work that was subsequently published and, among that, papers at major venues. From each paper’s title, abstract and introduction an LLM extracts innovation claims and assigns each an innovation aspect; a second LLM pass discards claims that are not innovation claims at all—open-source and reproducibility statements, bare state-of-the-

Table 4: Composition of IDEADELTA: counts of ideas and of target–prior pairs, and the idea-level novelty distribution, by source domain. Pairs are formed within a source domain only, so *All* is the union of the three subsets rather than a fourth collection.

	Graph ML	Computer vision	Time series	All
Ideas	5,225	8,291	7,595	21,111
Pairs	21,183	36,238	35,455	92,876
LOW	819	901	1,724	3,444
WEAK	1,240	1,711	1,614	4,565
MODERATE	2,247	2,777	2,173	7,197
HIGH	919	2,902	2,084	5,905

art assertions, and engineering deliverables—so that later stages judge substantive claims rather than boilerplate.

Candidate priors. Idea texts are embedded and indexed for top- K nearest-neighbour recall, and candidate relations are formed under a hybrid rule: high similarity is accepted directly, low similarity is rejected directly, and the uncertain band is passed to an LLM. The base build pairs ideas of the same type only, which would record an idea whose closest relative is of another type—a theoretical contribution nearest to a method—as isolated; a cross-type pass recovers these. At most $K = 5$ ranked priors are retained per target. Pairing stays within a source domain, and no cross-domain pair is labelled, so the three single-domain subsets and their union address the same ideas under identifiers namespaced by domain. The assembled dataset passes its own validation checks with no duplicate pair keys, no duplicate novelty targets, no invalid pair labels, no target left without a pair, and no temporal violation.

Annotation. All annotation stages use the same judge, `deepseek-v4-pro`. It labels pair coverage over the four classes, with the ambiguous band—related but not covering, partial cover, large cover, and high-similarity pairs the graph left unconnected—re-judged rather than accepted from the similarity rule. It then estimates, for each target, the substantive residual that survives the retained evidence, the sufficiency of that evidence, and the extent to which the claimed difference is surface-level. Idea-level grades follow by applying the novelty rule of Sec. 2.1 to those quantities. Labels are model-generated; each stage was reviewed by sampled manual inspection.

Composition. IDEADELTA contains 21,111 annotated ideas and 92,876 annotated target–prior pairs, with target dates from 2019-01 to 2026-09 (Table 4). Each idea carries its source domain, and the *All* column is the union of the three domain subsets rather than a fourth collection.

Splits. Splits are date-ordered rather than random, so that no training target is dated after an evaluation target. The cuts fall at 2025-02 and 2025-11, giving 13,815/3,944/3,352 ideas for training, validation, and test.

B.2 VALIDATION OF RESEARCH NOVELTY LABELS

Our novelty labels assess the substantive contribution of a target idea beyond its retrieved prior work. We validate this interpretation through complementary experiments. First, we examine how well metadata and retrieved prior content predict the existing labels. We then modify the evidence and repeat annotation to test whether novelty decreases when the target contribution is already covered and remains stable when its substantive contribution is unchanged.

Prediction from metadata. We first test whether simple metadata can account for the novelty labels without describing the technical content of prior work. We consider six variables: character length, word count, publication year, the innovation aspect and contribution type, and the number of retrieved priors. Over all 21,111 ideas, we measure their individual associations with the novelty grade using Spearman ρ for continuous variables and ε^2 for categorical variables. We set the thresholds in advance at $|\rho| \geq 0.20$ and $\varepsilon^2 \geq 0.06$.

Text length, word count, and publication year show weak associations ($\rho = +0.049, +0.058$, and -0.130 , respectively), while the innovation aspect and contribution type have $\varepsilon^2 \leq 0.033$. All remain below the corresponding thresholds. The number of priors and their mean coverage are negatively correlated with novelty ($\rho = -0.286$ and -0.471 , respectively). The negative association with coverage is consistent with the intended meaning of the labels: contributions already covered by prior work receive lower novelty grades.

Individual associations may overlook information available from several variables together. We therefore train a classifier on all six metadata variables to predict the four novelty grades. We use the date-ordered split of Appendix B.1, with 13,815 training ideas and 3,352 test ideas, and evaluate quadratic weighted kappa (QWK), which accounts for the ordering of the grades. For comparison, we train the same classifier using only mean prior coverage. QWK increases from 0.227 with metadata to 0.539 with coverage alone. Coverage thus predicts the labels more accurately than the metadata considered here, supporting the relevance of the relationship between the target and prior work.

Prediction with retrieved and random priors. We next examine whether the specific retrieved content matters for label prediction. In the TRUE condition, each target is paired with its five retrieved priors. In the RANDOM condition, these are replaced by five ideas sampled uniformly from those published before the target. The two conditions use the same number of priors and total evidence text volume, and all priors precede the target. We hold the model, hyperparameters, date-ordered split, and random seeds fixed. Models are trained separately from scratch and evaluated on the same held-out targets against their original labels.

Across the three seeds, QWK decreases from 0.196 with retrieved priors to 0.005 ± 0.021 with random priors. The gap of 0.192 is about an order of magnitude larger than the between-seed standard deviation of 0.015. Replacing retrieved priors with random earlier work therefore reduces prediction to near-chance agreement. This comparison supports the importance of the retrieved content, beyond the amount of evidence supplied.

Response to evidence covering the target contribution. The prediction experiments keep the labels fixed. We now test whether the annotation procedure responds appropriately when the evidence changes. If an earlier study already contains the target contribution, the assigned novelty should decrease. We sample 30 targets across the four grades and construct one synthetic prior for each target. This prior describes the same technical contribution and is dated before the target. After adding it to the evidence set, we repeat the complete annotation procedure: pair coverage labeling, residual contribution estimation, and grading according to the novelty rule of Sec. 2.1. Synthetic priors have dedicated identifiers and are marked `synthetic_control`. They do not refer to real papers, authors, or DOIs and are excluded from the released dataset.

Each modified target is compared with five independent annotations of its original evidence set to account for variation across repeated annotations. We evaluate recognition of the added prior as large coverage, changes in joint coverage and residual contribution, and decreases in the novelty score and grade. Ties are excluded from the exact sign test.

The added prior is labeled large coverage in all 29 comparable cases. Joint coverage increases, while residual contribution and novelty score decrease. The novelty grade decreases for 82.8% of targets (sign test $p = 2.2 \times 10^{-7}$). These results show that the annotation procedure lowers novelty when the evidence covers the target contribution.

Stability under changes that preserve the contribution. A novelty judgment should also remain stable when the evidence changes without altering the substantive contribution of the target. For the same 30 targets, we reorder the priors, duplicate existing priors, or add unrelated papers. The novelty grades before and after these changes show high agreement: $\kappa = 0.970, 0.902$, and 1.000 , respectively. These results complement the coverage experiment: the judgments respond to evidence that reduces the remaining contribution and remain comparatively stable under changes that preserve it.

Table 5: Symbols used in Stages A–C. Indices: k ranges over retained priors, m over semantic units, j over target tokens, and $i < j$ over unit pairs.

Symbol	Meaning	Size
<i>Sizes and index sets</i>		
d	encoder dimension	768
K	retained priors per target	5
M	semantic units	8
N	target tokens	—
n	retained priors, $ \mathcal{E}_x $	$\leq K$
$\mathcal{E}_x$	evidence set of target x	—
$\mathcal{V}$	non-padding target positions	—
<i>Stage A: historical evidence</i>		
π_k^{pair}	pair coverage distribution	$\mathbb{R}^4$
s_k	pair coverage score	$[0, 1]$
$\mathbf{f}_k$	pair coverage features	$\mathbb{R}^{11}$
$\mathbf{h}_k^J$	pair representation (joint model)	$\mathbb{R}^d$
$\mathbf{z}_J$	bag representation	$\mathbb{R}^d$
π^J	joint coverage distribution	$\mathbb{R}^4$
C_J	scalar joint coverage	$[0, 1]$
$\mathbf{z}_G$	global context	$\mathbb{R}^d$
<i>Stage B: residual decomposition</i>		
$\mathbf{v}_k, \tilde{\mathbf{v}}_k$	pair representation, augmented	$\mathbb{R}^d$
$\mathbf{h}_j$	target token state	$\mathbb{R}^d$
$\mathbf{q}_m$	learned unit query	$\mathbb{R}^d$
$\mathbf{u}_m$	semantic unit	$\mathbb{R}^d$
ϕ_m	unit activity	$[0, 1]$
A_{km}	support of prior k for unit m	$[0, 1]$
c_m	unit coverage (noisy-OR)	$[0, 1]$
r_m	unit residual weight	$[0, 1]$
$\hat{C}_U$	reconstructed joint coverage	$[0, 1]$
$\mathbf{t}_{ij}$	interaction representation	$\mathbb{R}^d$
g_{ij}	interaction strength	$[0, 1]$
$\bar{A}_{ij}$	co-support, $\max_k A_{ki} A_{kj}$	$[0, 1]$
r_{ij}	interaction residual weight	$[0, 1]$
η_{ij}	interaction attention	$[0, 1]$
$\mathbf{z}_U, \mathbf{z}_I, \mathbf{z}_C$	unit, interaction, covered pooling	$\mathbb{R}^d$
<i>Stage C: novelty prediction</i>		
$\mathbf{h}_x$	target representation	$\mathbb{R}^d$
γ	context gate	$[0, 1]$
$\mathbf{z}_R$	fused residual	$\mathbb{R}^d$
$\mathbf{h}$	shared representation	$\mathbb{R}^{4d}$

C METHOD DETAILS

C.1 NOTATION AND DIMENSIONS

Table 5 collects the symbols of all three stages. Every vector representation lies in $\mathbb{R}^d$, so the MLPs take inputs of dimension $4d+11$ for f_A , $4d$ for f_I and $3d$ for f_R , and the shared representation $\mathbf{h}$ has dimension $4d$. Section 3 defines the symbols themselves; the remainder of this appendix supplies the components each stage leaves unspecified.

C.2 HISTORICAL COVERAGE MODELS

Pair coverage and evidence selection. For each target–prior pair (x, p_k) , the pair coverage model estimates how much of the target the prior accounts for. A cross-encoder predicts relatedness a_k , coverage conditional on relatedness b_k , and large coverage conditional on coverage d_k , which factorize into a distribution over four ordered classes—unrelated, related but uncovered, partially covered,

and largely covered:

$$\pi_k^{\text{pair}} = [1 - a_k, a_k(1 - b_k), a_k b_k(1 - d_k), a_k b_k d_k]. \quad (1)$$

Indexing the classes from 0 to 3, the model scores the pair as $s_k = \frac{1}{2}\pi_{k,2}^{\text{pair}} + \pi_{k,3}^{\text{pair}}$, counting partial coverage at half the weight of large coverage. The pipeline ranks temporally admissible candidates by $s_k + 0.02\ell_k/3$, where ℓ_k is the predicted ordinal coverage label, and retains at most K of them. The retained priors and their pair coverage features form the evidence inputs to the subsequent stages.

Pair feature vector. The features carried into Stages B and C are

$$\mathbf{f}_k = [s_k; s_k + 0.02\ell_k/3; \pi_k^{\text{pair}}; a_k; a_k b_k; d_k; \kappa_k/K; \mathbb{K}[\text{prior text available}]] \in \mathbb{R}^{11}, \quad (2)$$

where κ_k is the rank of the prior within $\mathcal{E}_x$. Through Eq. (1), the first seven entries are deterministic functions of (a_k, b_k, d_k) , so $\mathbf{f}_k$ carries three degrees of freedom from the pair model alongside the two ranking quantities and the availability flag. We keep the redundant parameterization because the downstream projections consume these quantities at different scales.

Joint coverage. Different priors may cover different parts of the target, so a single pair score understates what the evidence establishes together. We therefore estimate joint coverage with a multiple-instance model that treats each pair (x, p_k) as an instance and the retained evidence set as a bag (Ilse et al., 2018). Let $n = |\mathcal{E}_x|$ denote the number of retained priors. A shared cross-encoder maps each pair to a representation $\mathbf{h}_k^J$, from which the model computes attention weights β_k and internal evidence scores e_k :

$$\beta_k = \text{softmax}_k(\mathbf{w}_a^\top \mathbf{h}_k^J + b_a), \quad e_k = \sigma(\mathbf{w}_e^\top \mathbf{h}_k^J + b_e). \quad (3)$$

For a nonempty evidence set, noisy-OR pools the evidence scores as $e_\cup = 1 - \prod_{k=1}^n (1 - e_k)$. Four statistics summarize the scores— that aggregate, their maximum and mean, and the fraction of the K evidence slots that are filled—and join the attention-pooled and elementwise maximum-pooled pair representations:

$$\mathbf{v}_E = \left[e_\cup, \max_k e_k, \frac{1}{n} \sum_{k=1}^n e_k, \frac{n}{K} \right], \quad (4)$$

$$\mathbf{z}_J = f_J \left(\left[\sum_{k=1}^n \beta_k \mathbf{h}_k^J; \max_k \mathbf{h}_k^J; \mathbf{v}_E \right] \right), \quad (5)$$

where f_J is an MLP. Attention pooling aggregates evidence across priors, while maximum pooling retains the strongest feature values; the noisy-OR aggregate enters only as one of the four statistics, and the prediction itself is learned from the full bag representation. A hierarchical head maps $\mathbf{z}_J$ to three probabilities: any coverage, substantial coverage conditional on any coverage, and mostly covered conditional on substantial coverage. Under the same factorization as Eq. (1), these probabilities yield π^J over four joint-coverage classes mirroring the pair classes: *not*, *weakly*, *partially*, and *mostly* covered. The scalar estimate is $C_J = \sum_{\ell=0}^3 \mu_\ell \pi_\ell^J$, where μ_ℓ is the mean annotated coverage score for class ℓ on the training split. Both C_J and π^J are supplied as fixed inputs to the residual and novelty modules.

Global context features. The twenty-one scalars projected into $\mathbf{z}_G$ are the scaled target times-tamp, ten pair-score statistics, and ten joint-coverage features. The pair-score statistics over $\mathcal{E}_x$ are the fraction of the K evidence slots that are filled; the maximum, mean, standard deviation and sum of the s_k ; a union score over them; the fractions of retained priors judged covering and judged related; the margin between the top two ranking scores; and the entropy of the pair scores. The joint-coverage features are two scalar coverage estimates, the raw and normalized entropies of π^J , an indicator for whether a joint prediction is available, the gap between the joint estimate and the pair-score union, and π^J itself.

C.3 RESIDUAL REPRESENTATIONS AND POOLING

This section specifies the encoders, pooling rules, and masking behavior that Section 3.2 leaves implicit.

Pair encoding. The residual cross-encoder shares parameters across all retained priors. For each target–prior pair (x, p_k) it produces token states $\mathbf{H}_k \in \mathbb{R}^{L \times d}$, where L is the input sequence length. Masked mean pooling over these states gives the pair representation, which we augment with projected pair coverage features:

$$\mathbf{v}_k = \text{MeanPool}(\mathbf{H}_k), \quad \tilde{\mathbf{v}}_k = \mathbf{v}_k + f_P(\mathbf{f}_k), \quad (6)$$

where f_P consists of a linear map, layer normalization, GELU, and dropout. The target token bank averages target-token states at matching positions across valid pairs, and masked mean pooling over the bank yields the target representation $\tilde{\mathbf{h}}_x$ that Stage C fuses.

Coverage reconstruction. To supervise the unit coverage estimates, we aggregate them into a scalar joint-coverage prediction. The pooling weights depend on both the semantic representation and the activity of each unit:

$$\hat{C}_U = \sum_{m=1}^M \omega_m c_m, \quad \omega_m = \frac{\phi_m \exp(\mathbf{w}_c^\top \mathbf{u}_m + b_c)}{\sum_{r=1}^M \phi_r \exp(\mathbf{w}_c^\top \mathbf{u}_r + b_c)}. \quad (7)$$

Annotated coverage and the fixed Stage A prediction supervise the resulting estimate $\hat{C}_U$. This supervision is aggregate: it constrains the unit coverages without determining the latent support matrix uniquely. Noisy-OR likewise serves as a differentiable rule for aggregating support, and using it does not establish that support from different priors is statistically independent.

Interaction encoding and attention. For each unit pair $i < j$, an MLP f_I encodes the two units and their elementwise comparisons into an interaction representation. A sigmoid head then predicts the interaction strength:

$$\mathbf{t}_{ij} = f_I([\mathbf{u}_i; \mathbf{u}_j; \mathbf{u}_i \odot \mathbf{u}_j; |\mathbf{u}_i - \mathbf{u}_j|]), \quad g_{ij} = \sigma(\mathbf{w}_g^\top \mathbf{t}_{ij} + b_g). \quad (8)$$

Using the interaction residual weights r_{ij} defined in Stage B, we compute attention over all $\binom{M}{2}$ unit pairs:

$$\eta_{ij} = \text{softmax}_{i < j} (\mathbf{w}_I^\top \mathbf{t}_{ij} + b_I + \log(r_{ij} + \varepsilon)), \quad \mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}. \quad (9)$$

The logarithmic bias directs attention toward pairs with larger residual weights. Multiplying by r_{ij} after normalization then restores their scale, so that weak residuals contribute less than the softmax alone would allow—the same reason $\mathbf{z}_U$ is normalized by M rather than by $\sum_m r_m$.

Masking and edge cases. Padded evidence slots carry zero support: the evidence-availability flag masks the support matrix, which is clamped to $[0, 0.995]$ before use. Both aggregations remain well defined when fewer than K priors are retained. The noisy-OR product runs over the retained priors only, while the co-support maximum is taken over all K slots, so $\hat{A}_{ij} = 0$ when none is retained—an empty record yields no evidence that the combination exists.

C.4 RESIDUAL FUSION, PREDICTION HEADS, AND DECISION RULES

This section specifies the gate, the heads, and the decision rules that Section 3.3 states only in outline.

Context gate. The fusion gate conditions the contribution of global context on the two residual representations and the uncertainty of the joint-coverage prediction. Let H_J denote the normalized entropy of π^J . The gate is

$$\gamma = \sigma(f_\gamma([\mathbf{z}_U; \mathbf{z}_I; \mathbf{z}_G; H_J])). \quad (10)$$

Here f_γ is a two-layer MLP with a $d/2$ bottleneck and a scalar output, so γ scales $\mathbf{z}_G$ uniformly rather than gating it per dimension. The gated context $\gamma \mathbf{z}_G$ enters residual fusion alongside $\mathbf{z}_U$ and $\mathbf{z}_I$, while $\mathbf{z}_G$ reaches the shared representation—and through it the prediction heads—on an ungated path of its own.

Auxiliary predictions. The auxiliary heads use the shared representation $\mathbf{h}$ to predict substantive difference $\hat{d}$, surface change $\hat{s}$, evidence sufficiency $\hat{e}$, and residual-type probabilities π^T . The first three outputs are sigmoid scores; the type prediction uses the hierarchy described below.

Residual types. The type hierarchy distinguishes a minor variant from six substantive families: conceptual innovation, mechanism or system, task or application, training or evaluation, data-related innovation, and combination. Conditional fine heads yield eleven types in total. The minor/substantive gate additionally reads $\hat{d}$, $\hat{s}$, and $\max_{i < j} r_{ij}$. The probability assigned to the combination type is multiplied by $0.70 + 0.30\sigma(f_{\text{comb}}(\mathbf{z}_I))$ before the hierarchical distribution is renormalized. The final type distribution is a convex mixture of the hierarchical and flat predictions, with flat weight 0.15.

Novelty branches. All novelty branches receive the prediction features

$$\xi = [\mathbf{h}; \hat{d}; \hat{s}; \hat{e}; \pi^T; C_J; \pi^J]. \quad (11)$$

We encode LOW, WEAK, MODERATE, and HIGH as 0, 1, 2, and 3, respectively. A direct linear classifier and a two-layer MLP classifier map ξ to class distributions $\mathbf{p}^D$ and $\mathbf{p}^R$.

The ordinal branch models cumulative probabilities of reaching each grade threshold. It produces three sigmoid gates g_1, g_2, g_3 from ξ and defines

$$q_1 = g_1, \quad q_2 = g_1 g_2, \quad q_3 = g_1 g_2 g_3, \quad \mathbf{p}^O = [1 - q_1, q_1 - q_2, q_2 - q_3, q_3]. \quad (12)$$

Here q_ℓ estimates the probability that $y \geq \ell$, with $q_1 \geq q_2 \geq q_3$ by construction.

Probability fusion. We aggregate the three branch distributions through a convex mixture:

$$\mathbf{p} = \lambda_D \mathbf{p}^D + \lambda_O \mathbf{p}^O + \lambda_R \mathbf{p}^R, \quad \lambda_D + \lambda_O + \lambda_R = 1, \quad \lambda_D, \lambda_O, \lambda_R \geq 0. \quad (13)$$

The fused distribution provides the initial grade for the refinement below.

Moderate-to-high refinement. A separate binary head $h_{MH} = \sigma(f_{MH}(\xi))$ scores the distinction between moderate and high novelty. At inference, let $y_0 = \arg \max_\ell p_\ell$ for the mixture in Eq. (13), and define

$$h_{\text{high}} = \alpha_D p_3^D + \alpha_O p_3^O + \alpha_{MH} h_{MH}, \quad \hat{y} = \begin{cases} 3, & y_0 = 2 \text{ and } h_{\text{high}} \geq \theta_{MH}, \\ y_0, & \text{otherwise.} \end{cases} \quad (14)$$

where $\alpha_D, \alpha_O, \alpha_{MH} \geq 0$ and $\alpha_D + \alpha_O + \alpha_{MH} = 1$. Thresholding the weighted score, rather than requiring separate agreement from all three heads, promotes a MODERATE prediction to HIGH whenever the heads are collectively confident—even when each of them favors a different grade. We select the mixture weights and the refinement threshold on development data and fix them for testing; Appendix C.5 reports the values.

Continuous outputs. A separate sigmoid head predicts a continuous novelty score. Probability fusion and the moderate-to-high refinement alone determine the grade at inference, so this score never enters the decision rule. It is also a different readout from the substantive-difference head, which the blockwise RINoBench evaluation uses.

C.5 TRAINING AND IMPLEMENTATION DETAILS

This section defines the loss terms of the training objective and reports the settings of the residual stage.

Objective. Let $\mathcal{A} = \{\Delta, S, E, C, \nu\}$ index the annotated scalars—the substantive difference, surface change, and evidence sufficiency read off the auxiliary heads, the reconstructed coverage $\hat{C}_U$, and the continuous novelty score. Each contributes a smooth- L_1 level loss $\mathcal{L}_a$ against its annotation, and the ranking terms of Δ, S, E and C are collected into a single $\mathcal{L}_{\text{rank}}$, the novelty score carrying its own inside $\mathcal{L}_\nu$:

$$\mathcal{L} = \sum_{a \in \mathcal{A}} \lambda_a \mathcal{L}_a + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{nov}} \mathcal{L}_{\text{nov}} + \lambda_{\text{type}} \mathcal{L}_{\text{type}} + \lambda_{\text{dec}} \mathcal{L}_{\text{dec}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}}, \quad (15)$$

Here $\mathcal{L}_{\text{nov}}$ supervises the three novelty branches, their fused distribution, and the moderate-to-high head; $\mathcal{L}_{\text{type}}$ supervises residual categories and nontrivial combinations; and $\mathcal{L}_{\text{dec}}$ and $\mathcal{L}_{\text{var}}$ regularize the decomposition. Table 6 lists every coefficient λ .

Supervision. $\mathcal{L}_{\text{nov}}$ trains the direct and MLP classifiers with class-weighted cross-entropy, the ordinal outputs against $\mathbf{1}[y \geq \ell]$, $\ell = 1, 2, 3$, with weighted binary cross-entropy, and the fused distribution with a negative log-likelihood. The moderate-to-high head sees only moderate and high examples and adds a positive-negative margin to its binary cross-entropy, weighting moderate examples scored as high more heavily. $\mathcal{L}_{\text{type}}$ combines the hierarchy’s conditional losses, a flat classification loss, binary supervision for nontrivial combinations, and consistency penalties tying the combination branch to the type predictions. The novelty score regresses ν^* , the continuous target that $\text{raw}(x)$ in the novelty rule of Sec. 2.1 forms from the annotated coverage, substantive difference, evidence sufficiency and surface change of the target, and that the dataset thresholds into the four grades.

Scalar heads: level and ranking terms. Let ℓ_δ denote smooth- L_1 loss and w_x the example weight. For $a \in \mathcal{A}$, the level loss is $\mathcal{L}_a = \sum_x w_x \ell_\delta(\hat{a}_x - a_x^*) / \sum_x w_x$, where a_x^* is the annotation and $\hat{a}_x$ the prediction. The ranking term takes within-minibatch pairs separated by $|a_x^* - a_{x'}^*| \geq 0.05$ and contributes $[0.02 - \text{sgn}(a_x^* - a_{x'}^*)(\hat{a}_x - \hat{a}_{x'})]_+$, weighted by $\sqrt{w_x w_{x'}}$; it is dropped when a minibatch holds no eligible pair.

Ranking is not redundant with the level loss. At the transition δ and target spread used here the level loss operates in its L_1 regime, whose optimum is the median of the target, so a head supervised by it alone drifts towards a near-constant output—the evidence head most visibly. Ranking is scale-free and counteracts that drift, which is also why it takes most of the weight inside $\mathcal{L}_\nu$: thresholding reads the grade off the score, so only order matters there.

Coverage, decomposition, and variance terms. Beyond its level loss against the annotated joint value C^* , $\mathcal{L}_C$ adds a consistency loss against the upstream score C_J at relative weight 0.2, reweighted by $1 - H_J$ so that examples with uncertain upstream coverage count for less; no unit-level support matrix is supplied as a target. $\mathcal{L}_{\text{dec}}$ combines three terms: attention regularization penalizes the squared entries of $\tilde{\alpha}\tilde{\alpha}^\top - I$ for the row-normalized attention matrix $\tilde{\alpha}$, plus half the mean off-diagonal cosine similarity; query diversity penalizes the mean squared cosine similarity between distinct queries; and a usage penalty $D_{\text{KL}}(\rho \| \mathbf{u})$, with $\rho_m = \sum_j B_{mj} / \sum_{r,j} B_{rj}$ and $\mathbf{u}$ uniform over the M units, keeps units from going unused. The variance floor carries its own coefficient,

$$\mathcal{L}_{\text{var}} = \mathbb{E}_x [\nu - \text{Var}_m(c_{x,m})]_+ / \nu, \quad (16)$$

and acts on the unit coverages rather than on the token assignments, keeping the c_m within an example from coinciding.

Curriculum and feature perturbation. For a randomly sampled fraction of training examples, values derived from annotated joint coverage replace the joint-coverage features and the associated probability and entropy features—with probability 0.50 in epoch 1, 0.20 in epoch 2, and 0 thereafter. Development and test runs always use the upstream joint predictions. Selected joint-coverage features on the global-context path also receive Gaussian noise of standard deviation 0.03 and independent feature dropout at probability 0.10, which leaves untouched both the direct coverage inputs to the novelty heads and the entropy input to the fusion gate.

Inputs and initialization. The input adapter reconstructs pair probabilities missing from a ranking record from the available coverage label and score; pair labels and scores also appear in the pair text input. The two facet embeddings entering z_G have dimension 32 each. The model uses DeBERTa-v3-base with hidden dimension 768, maximum pair-input length 384, $K = 5$ and $M = 8$; the residual encoder is initialized from the learned joint model, and upstream checkpoints stay fixed. Padded priors have zero support, support values are capped at 0.995, and an empty evidence set falls back to a target-containing input with all prior support masked out. Training uses refinement weights $(\alpha_D, \alpha_O, \alpha_{MH}) = (0.40, 0.30, 0.30)$ and mixture weights $(\lambda_D, \lambda_O, \lambda_R) = (0.50, 0.35, 0.15)$; development-set calibration selects $(0.60, 0.20, 0.20)$ and $\theta_{MH} = 0.32$ for inference, fixed thereafter. Table 6 lists the remaining settings.

Table 6: Residual-stage settings for the reported IdeaDelta checkpoint. The checkpoint was selected on development data at epoch 5 of 8.

Setting	Value
Training batch size / evaluation batch size	2 / 8
Head learning rate / encoder learning rate	$1.5 \times 10^{-5} / 3 \times 10^{-6}$
Weight decay / dropout	0.01 / 0.15
Smooth- L_1 transition δ	0.25
$\lambda_\Delta / \lambda_S / \lambda_E$	1.0 / 0.6 / 0.8
λ_C / λ_ν	0.3 / 0.6
$\lambda_{\text{nov}} / \lambda_{\text{type}}$	0.8 / 1.0
λ_{rank}	0.5
$\lambda_{\text{dec}} / \lambda_{\text{var}}$	0.05 / 0.02
Within $\mathcal{L}_{\text{rank}}$: $\Delta / S / E / C$	1.0 / 0.5 / 0.5 / 0.5
Within $\mathcal{L}_\nu$: ranking weight	2.0
Within $\mathcal{L}_C$: Stage A consistency weight	0.2
Within $\mathcal{L}_{\text{dec}}$: attention / query / usage	0.7 / 0.1 / 0.2
Variance floor ν	0.0025
Seed	173

D EXPERIMENTAL DETAILS

D.1 SETUP

Splits. The split is a date-ordered 13,815/3,944/3,352 partition of the 21,111 idea-level samples in IDEADELTA.

Configuration. All runs share one configuration and vary a single parameter at a time: deberta-v3-base backbone, 8 epochs, batch size 2, learning rate 1.5×10^{-5} with an encoder learning-rate ratio of 0.2, eight semantic units with slot competition enabled, and a fixed joint-prediction checkpoint supplying the encoder initialisation. The checkpoint reported in the main text is selected on dev; test is traversed once, at final evaluation, and is not used to select a checkpoint.

Per-domain metrics. Per-domain figures are recomputed from the stored predictions rather than read from the trainer, and reconcile with the trainer’s own pooled figures to within 5×10^{-5} on all six metrics.

D.2 PARAMETER SENSITIVITY

Each sweep varies one parameter from the default configuration of Table 1 and holds the rest fixed: the upstream pair and joint products, the encoder initialisation, the evaluation files and the seed. Every point is a single run at seed 173, except for the seed sweep below.

Evidence budget: the ideas run out of priors before the model runs out of capacity. Figure 2 varies K , the number of priors each idea contributes, over $K = 2, 3, 4, 5, 6, 8$. Accuracy rises steeply from $K=2$ to $K=4$ (0.731 $\rightarrow$ 0.806 on Graph ML, 0.716 $\rightarrow$ 0.772 on computer vision, 0.726 $\rightarrow$ 0.789 on time series) and then flattens, with the same knee in all three domains. The flattening comes from the evidence rather than from the model. Because every sample is padded to exactly K slots, the priors a setting consumes stop growing once K exceeds the candidates an idea carries, and the knee sits at three to four consumed priors. Raising K from 5 to 8 adds less than one consumed prior, because about a third of the ideas carry six or more candidates and fewer than one in seven carry eight or more.

Training-set size: more data helps across the board. Figure 3 varies the number of training ideas over 25%, 50%, 75% and 100% of the training split, drawn as nested stratified prefixes so that consecutive points add samples without reordering what came before. Macro-F1 rises steadily with the fraction on Graph ML (0.7627 $\rightarrow$ 0.8010) and time series (0.7573 $\rightarrow$ 0.7760). Computer vision

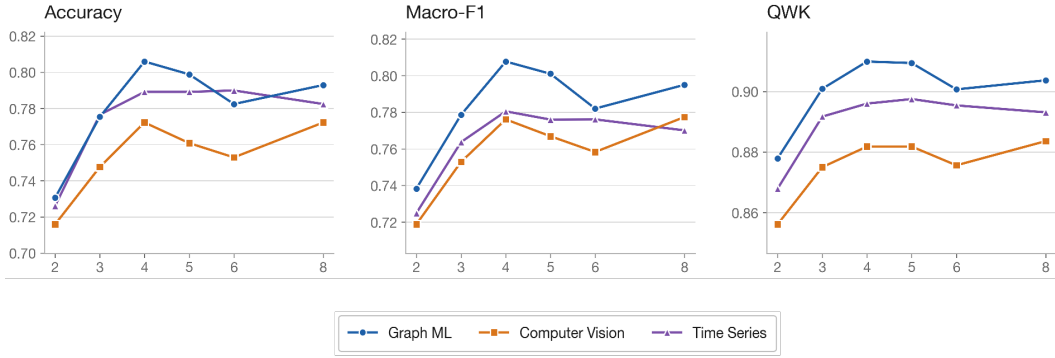


Figure 2: Sensitivity to the evidence budget K in the three domains. The default is $K=5$, the checkpoint of Table 1. K is a padding dimension, so the text reads the curves against the priors each setting consumes.

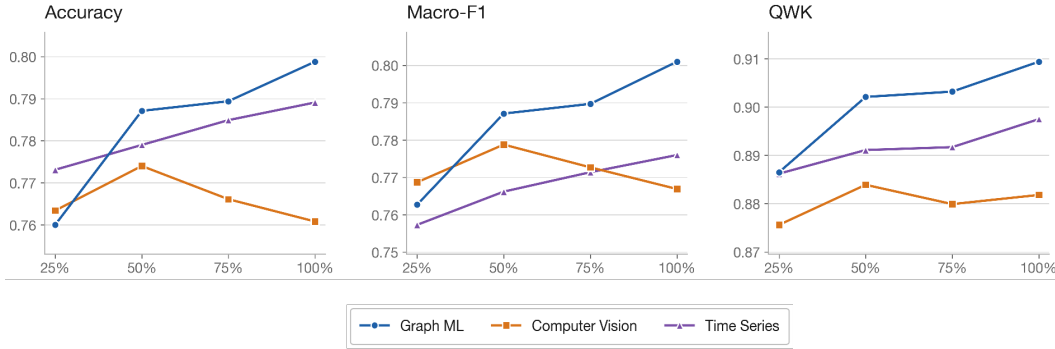


Figure 3: Sensitivity to the amount of EGRD training data in the three domains. Development and test sets stay fixed, and 100% is the default.

levels off after 50% and ends at 0.7669 against a peak of 0.7788, a gap of 0.012. More training data helps in most of the domains we measure, and the spread across the four fractions stays small: accuracy moves by at most 0.039 in any domain.

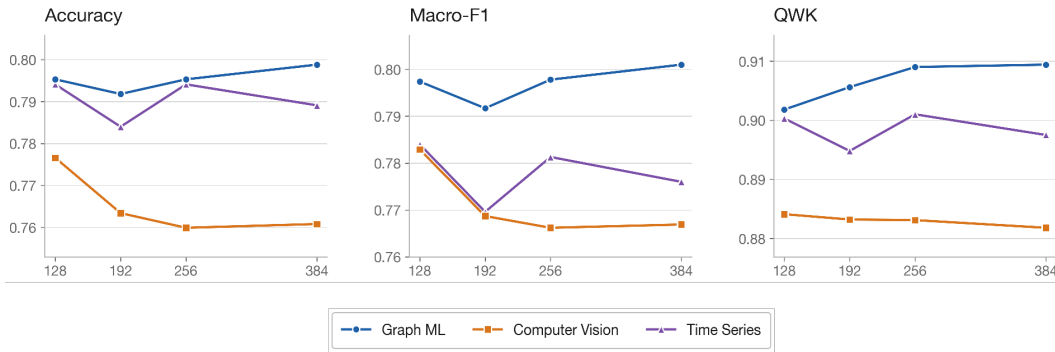


Figure 4: Sensitivity to encoder input length on the three domain test splits. The default is 384, and every setting scores the same test ideas and labels.

Encoder input length: 128 tokens already cover the comparison. Figure 4 varies the encoder input length over 128, 192, 256 and 384 tokens. The four lengths separate the domains by little: no domain moves monotonically with input length, and tripling the input moves Macro-F1 by at most 0.017 in any domain. At 128 tokens the encoder already covers the title-and-abstract comparison for

most target–prior pairs, so a longer window reads much the same text and the curves stay flat. QWK is the flattest panel, spanning 0.0076 on Graph ML, 0.0023 on computer vision and 0.0062 on time series across the four lengths, so the ordering the model produces is close to invariant to how much of the text it reads. The flatness is consistent with the decomposition reading a target and its priors through short unit-level comparisons rather than through long spans of text.

Table 7: Per-domain seed variability. Entries are sample standard deviations across seeds 173, 174 and 175 with identical data, code and configuration.

Domain	Acc	Ma-F1	QWK	MAE	High-F1
Graph ML	0.0066	0.0050	0.0048	0.0067	0.0022
Computer vision	0.0040	0.0021	0.0032	0.0027	0.0037
Time series	0.0050	0.0045	0.0031	0.0051	0.0069

Seed stability: the reported model is not a selected seed. Table 7 varies nothing but the random seed, running the default configuration on seeds 173, 174 and 175 with identical data and code and reporting the per-domain sample standard deviations. The spread is 0.004–0.007 in accuracy and 0.002–0.005 in Macro-F1, so the headline numbers describe the configuration rather than one fortunate run, and the three sweeps above should be read by the shape of a curve rather than by the difference between adjacent points.

E WHAT THE REPRESENTATION ORGANISES

The grade our model predicts and the representation it predicts it from are not the same object, and the second is the more informative one to inspect. This section reads six stage representations out of a single forward pass of one frozen checkpoint over the 3,352 test ideas: the target encoding, the covered pathway z_C , the two residual pathways z_U and z_I , the global feature context, and the fused vector the prediction heads receive.

Choosing a target the input scalars cannot reach. Two quantities enter the model as features and between them carry most of the four-way novelty grade: the joint coverage of the retrieved priors and the number of admissible candidates. A multinomial probe on those two scalars and their low-order interactions reaches 0.681 accuracy on the grade, against a 0.329 majority baseline and the model’s own 0.778. Any structure we find by probing for the grade would therefore be reporting, in large part, on those two numbers.

The residual type does not have this problem. It is an eleven-way label naming what kind of contribution remains once the priors are accounted for — a new mechanism, a non-trivial combination, a minor variant, a new evaluation protocol, and so on — and it is not a function of how much coverage there is. The same two-scalar probe reaches only 0.368 on it against a 0.275 majority, while the model’s own head reaches 0.651. The gap of twenty-eight points is the part that had to be read out of the text, and it is what the rest of this section examines. Figure 5 collects the result.

The types are laid out, and the layout predicts the model’s own errors. Projecting the eleven centroids to two dimensions recovers 61.9% of their spread, so the arrangement is close to planar. Its first axis correlates at -0.905 with the mean novelty grade of each type across the eleven classes, which places the conceptual and task-defining contributions at one end and minor variants and recombinations at the other.

The arrangement is not decoration: it predicts where the model goes wrong. Over the 55 type pairs, the cosine between two centroids correlates at $r = 0.469$ with how often the model swaps those two types. The model never mistakes a new data distribution for a new theoretical concept, the most distant pair in the map, and its single most frequent confusion, minor variant against non-trivial combination, is one of its closest.

That correlation could be a restatement of the novelty axis, since types of similar novelty would be confusable for that reason alone. It is not. Partialling out the difference in mean novelty between the two types raises the correlation to 0.600 rather than removing it. The clearest case is

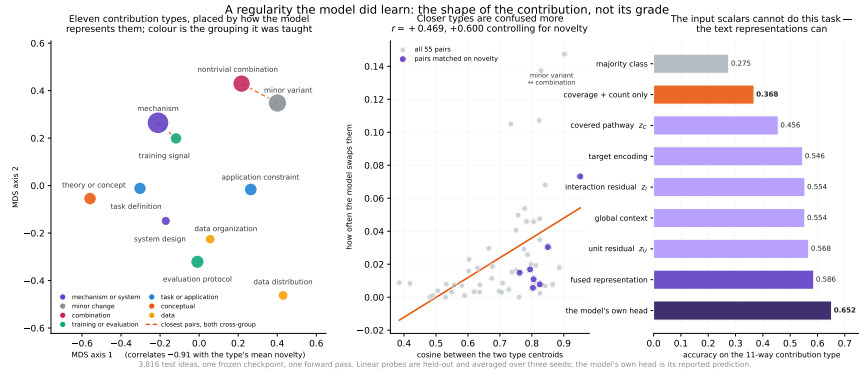


Figure 5: Left: the eleven contribution types, placed by classical multidimensional scaling on the cosine distances between their centroids in the fused representation; two dimensions carry 61.9% of the centroid spread, and point area is class frequency. Colour is the seven-way coarse grouping the model is supervised with, so same-coloured points sitting apart means the layout is not a replay of that supervision; the dashed segments mark the two closest pairs in the whole map, both of which cross groups. Centre: each of the 55 type pairs, its centroid cosine against how often the model swaps the two. Right: held-out accuracy of a linear probe for the eleven-way type, averaged over three seeds.

NEW_MECHANISM against NEW_TRAINING_SIGNAL, whose mean grades differ by 0.05 and which novelty therefore cannot separate at all: they are the closest pair in the entire map at cosine 0.951, and among the novelty-matched pairs they are also the most often confused.

The layout disagrees with the grouping the model was taught. Training supervises a seven-way coarse grouping alongside the fine label, so the layout could be that grouping made visible. It is not. Pairs inside a coarse group sit at mean cosine 0.783 and pairs across groups at 0.709, a separation far smaller than the spread within either set, and the four closest pairs in the map all cross groups. The model places MINOR_VARIANT beside NONTRIVIAL_COMBINATION at 0.902 although the supervision assigns them to different coarse types, and it pulls NEW_MECHANISM away from NEW_SYSTEM_DESIGN to 0.798 although the supervision groups them together. Both departures read as the more natural organisation: a minor variant and a recombination are two degrees of reusing what exists, whereas a new mechanism and a new system design are different kinds of contribution. We report this as an observation about the representation, not as evidence that the taxonomy should be redrawn; with only four same-group pairs available the comparison of the two means is descriptive.

Where the knowledge sits. Probing each stage separately for the eleven-way type puts the covered pathway lowest at 0.456, the target encoding at 0.546, the interaction residual at 0.554, the global context at 0.554, the unit residual highest among the single pathways at 0.568, and the fused vector at 0.586. Two readings follow. Every text-derived representation is well clear of the 0.368 the input scalars reach, so the contribution type is genuinely carried by the encoded text rather than by the retrieval statistics. And the target encoding alone already reaches 0.546, which says that much of the type is legible from how the idea is phrased — a target that says it combines two things announces its own category — so the evidence pathways add less here than the gap to the model’s 0.651 head might suggest.

F RINOBENCH EVIDENCE BUDGET

The blockwise rule lets the frozen checkpoint, trained with at most five candidate priors per idea, read more evidence than it ever saw in training by running once per block of five and taking the minimum. Raising the budget from 25 priors to 60 moves accuracy from 0.4116 to 0.4152 and QWK from 0.2182 to 0.2245. The gain is real but small, and it saturates: RINoBench ideas carry a median of 23 candidate priors, so beyond roughly 45 the later blocks cover too few ideas to matter.

1188 A thirteenth block was not run because only seven test ideas still carry a prior at that depth and the
1189 join against the joint-coverage stage returns empty.
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